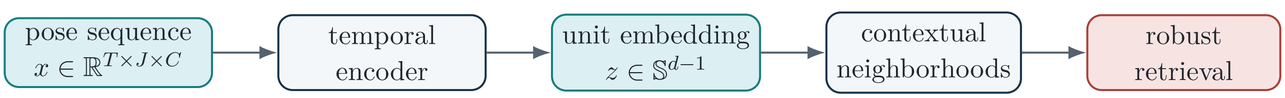
Contextual Similarity Optimization for Robust Pose-Sequence Retrieval

A mathematical and implementation guide to
Liao, Tsiligkaridis, and Kulis (ICML 2023)



Audience. Assumes basic probability, statistics, and machine learning. It develops the paper's equations from first principles, derives the neighborhood objective, explains the non-standard gradients, and maps the method into a controlled pose-retrieval experiment.

Prepared for today's final-proposal discussion
July 2026

How to use this guide today

The 12-minute route

Read these pages in order before the meeting:

1. **The one-sentence research claim** on this page.
2. **The full method in one picture** in figure 3.1.
3. **The five equations you must be able to narrate**: equations (1.7) and (P2) to (P6).
4. **Why this is a valid ranking objective**: section 8.1.
5. **The implementation plan**: chapter 12.
6. **The discussion script and likely questions**: chapter 14.

The one-sentence research claim

Test whether a neighborhood-aware metric-learning objective preserves the ranking quality of pose-sequence embeddings better than standard objectives when query skeletons are jittered, incomplete, temporally damaged, or produced by a shifted pose estimator.

The contextual paper demonstrates robustness to *label noise and overfitting in image retrieval*. It does **not** establish robustness to corrupted pose coordinates. The proposed project is compelling precisely because that transfer is an open, falsifiable hypothesis rather than a guaranteed consequence.

Equations marked **P1–P31** reproduce or algebraically restate the numbered equations in Liao, Tsiligkaridis, and Kulis. Motion-specific equations, implementation recommendations, and robustness metrics are explicitly labeled as adaptations or proposed extensions. One line in the paper's printed Algorithm 1 appears inconsistent with its Eq. (5): the algorithm line shows an MSE between contextual and cosine similarity, while Eq. (5), the proof, and the released code optimize contextual similarity against the binary label-similarity matrix. This guide follows Eq. (5) and the reference implementation.

Contents

List of Figures

1

Retrieval Geometry from First Principles

1.1 Retrieval is a ranking problem

A retrieval system receives a query q and ranks a gallery $\mathcal{G} = \{g_1, \ldots, g_m\}$. The output is not a single class probability. It is an ordering

$$g_{\pi(1)}, g_{\pi(2)}, \ldots, g_{\pi(m)}$$

chosen so that relevant items appear before irrelevant items. For class-based motion retrieval, relevance can begin with

$$\mathrm{rel}(q, g_j) = \mathbb{1}[y_q = y_j].$$

This is deliberately coarse, but objective and reproducible.

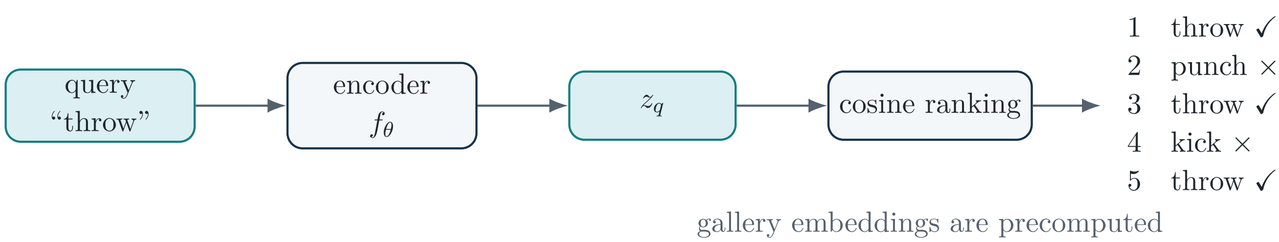


Figure 1.1: A retrieval system learns an ordering, not merely a class decision.

1.2 Embeddings and normalization

Let an encoder map an input x_i to an unnormalized vector $h_i = f_\theta(x_i) \in \mathbb{R}^d$. The paper works with unit-normalized features

$$z_i = \frac{h_i}{\|h_i\|_2}, \qquad \|z_i\|_2 = 1. \tag{1.1}$$

The unit sphere removes vector magnitude as a degree of freedom. The model can encode similarity only through direction.

The cosine similarity is therefore simply a dot product:

$$s_{ij} = \langle z_i, z_j \rangle \in [-1, 1]. \tag{1.2}$$

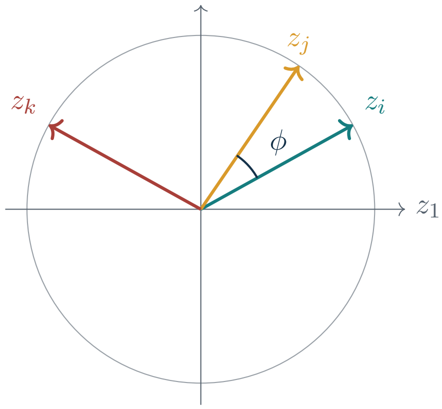


Figure 1.2: Unit-normalized embeddings convert cosine similarity into geometry on a sphere.

1.3 Why squared Euclidean distance becomes $2 - 2s_{ij}$

The paper switches between cosine similarity and squared Euclidean distance. This identity is central:

Because $\|z_i\| = \|z_j\| = 1$,

$$\begin{aligned} D(i,j) &= \|z_i - z_j\|_2^2 && (1.3)\\ &= \langle z_i - z_j, z_i - z_j\rangle && (1.4)\\ &= \|z_i\|^2 + \|z_j\|^2 - 2\langle z_i, z_j\rangle && (1.5)\\ &= 1 + 1 - 2s_{ij}. && (1.6)\end{aligned}$$

Therefore

$$\boxed{D(i,j) = 2 - 2s_{ij} \in [0,4].} \tag{1.7}$$

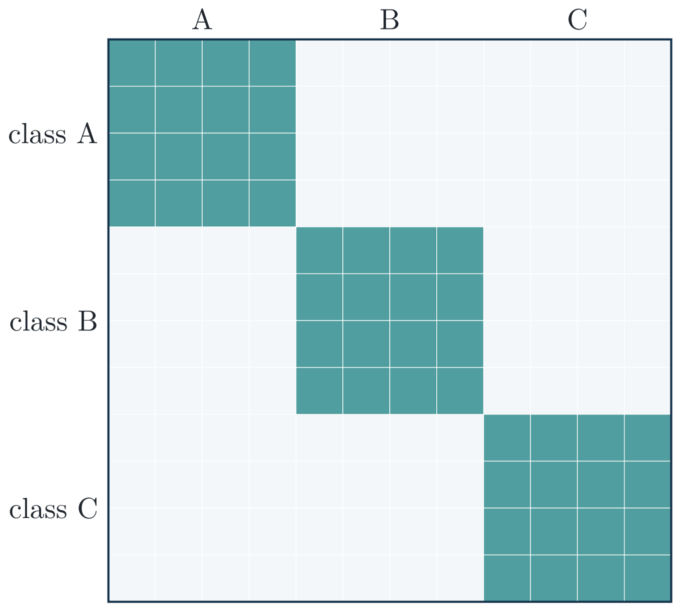
Ranking by decreasing cosine similarity is exactly the same as ranking by increasing squared Euclidean distance.

1.4 The label-similarity matrix

For a mini-batch of n labeled examples, define

$$y_{ij} = \mathbb{1}[y_i = y_j] \in \{0,1\}. \tag{1.8}$$

The matrix $\mathbf{Y} = [y_{ij}]$ is the target pattern. Under class-balanced ordering, it has block-diagonal structure.



teal: $y_{ij} = 1$
white: $y_{ij} = 0$

Figure 1.3: The binary target similarity matrix for a batch with three classes and four samples per class.

What to say

"The network outputs unit vectors. Cosine similarity gives the ranking, and because the vectors are normalized, squared Euclidean distance is just $2 - 2$ times the cosine similarity. The loss therefore changes one pairwise similarity matrix, but it does so through neighborhood structure rather than only individual pairs."

2

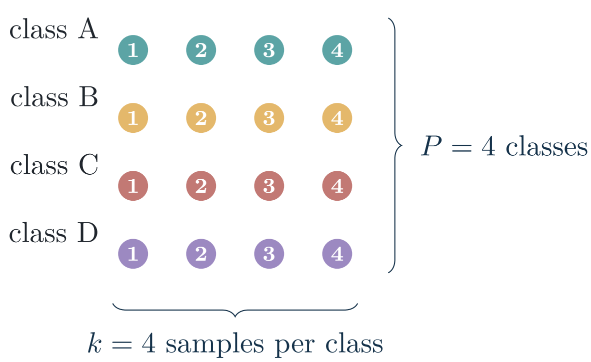
Metric-Learning Baselines and Balanced Batches

2.1 Why batch composition is part of the method

The paper uses a $P \times K$ sampler: choose P classes and K samples per class. To match the paper's notation, this guide writes

$$k = K, \qquad n = Pk.$$

The contextual neighborhood size is **not a free top-k tuning parameter**: the paper sets it equal to the number of samples from each class in the mini-batch.



$$n = Pk = 16$$

Figure 2.1: A balanced batch is a structural assumption, not a minor dataloader choice.

Why k equals samples per class

If ranking is perfect inside a balanced batch, each sample's k nearest items, counting itself, are exactly the k examples with the same label. This identity is what lets zero contextual loss coincide with correct within-batch ranking in the paper's proposition.

2.2 Pairwise contrastive loss

The contextual framework retains a standard pairwise term. In similarity form, the paper uses positive and negative margins δ_+ and δ_-:

$$\begin{aligned}\mathcal{L}_{\text{contrast}} = &\frac{\sum_{i,j:y_{ij}=1}[\delta_+ - s_{ij}]_+}{|\{(i,j) : y_{ij} = 1,\ \delta_+ - s_{ij} > 0\}|} \\ &+ \frac{\sum_{i,j:y_{ij}=0}[s_{ij} - \delta_-]_+}{|\{(i,j) : y_{ij} = 0,\ s_{ij} - \delta_- > 0\}|}.\end{aligned} \tag{P29}$$

Here $[a]_+ = \max(a, 0)$. A positive pair is penalized if its similarity is below δ_+; a negative pair is penalized if its similarity is above δ_-. Only margin-violating pairs contribute.

2.3 Triplet and supervised contrastive baselines

These are not part of the contextual paper's final formula, but they are critical controls for the proposed motion study.

For an anchor a, positive p, negative n, triplet loss is

$$\mathcal{L}_{\text{triplet}} = [D(a,p) - D(a,n) + m]_+. \tag{2.1}$$

It asks only for a relative inequality.

For supervised contrastive learning, define $P(i)$ as the set of same-label positives for anchor i and $A(i)$ as all other batch indices. With temperature τ,

$$\mathcal{L}_{\text{supcon}} = \sum_i -\frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i^\top z_p/\tau)}{\sum_{a \in A(i)} \exp(z_i^\top z_a/\tau)}. \tag{2.2}$$

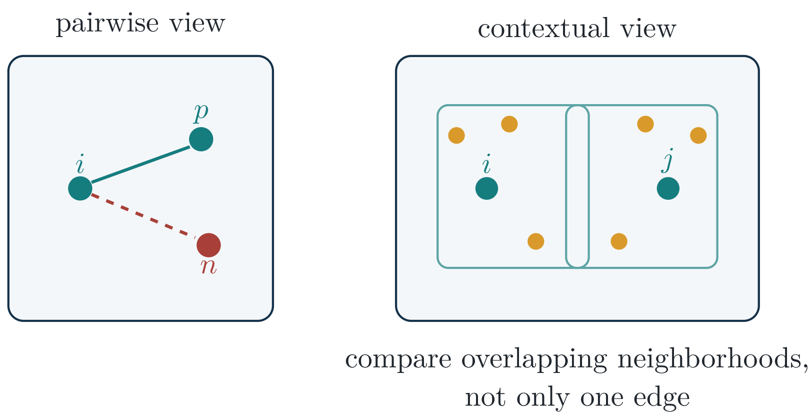


Figure 2.2: Standard losses reason through pairs or triplets; contextual loss also reasons through the local graph around each sample.

3

The Full Contextual-Similarity Pipeline

3.1 One picture before the details

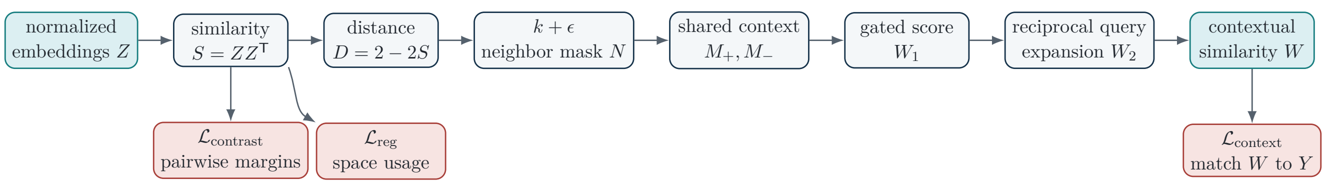


Figure 3.1: The complete training computation. Every contextual quantity is ultimately a function of the cosine-similarity matrix.

3.2 Dimensions and tensor meanings

Symbol	Shape	Meaning
Z	$n \times d$	one normalized embedding per mini-batch item
S	$n \times n$	cosine similarity, $S_{ij} = z_i^\top z_j$
D	$n \times n$	squared Euclidean distance, $D = 2\mathbf{1}\mathbf{1}^\top - 2S$ entrywise
Y	$n \times n$	binary label agreement
N	$n \times n$	differentiable-in-backward binary neighborhood indicator
M_+, M_-	$n \times n$	shared-neighbor and shared-non-neighbor counts
W_1, W_2, W	$n \times n$	progressively refined contextual similarities

The conceptual compression

The entire paper can be remembered as

$$\boxed{\text{embeddings} \to \text{pairwise similarities} \to \text{local graph} \to \text{graph similarity} \to \text{label graph.}}$$

The difficult part is making the discrete local graph trainable.

4

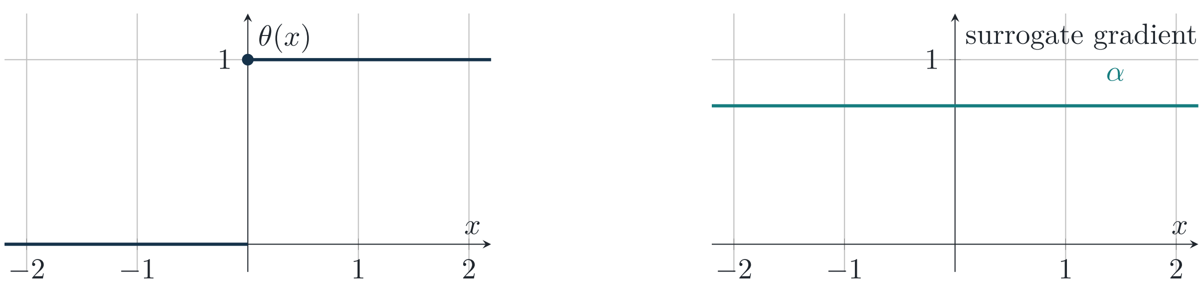
Step 1: Differentiable Neighborhoods

4.1 The Heaviside operation and the heuristic gradient

A top-k neighborhood is discrete. The paper expresses the threshold with a Heaviside function:

$$\text{Forward:}\quad \theta(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0, \end{cases} \qquad \text{Backward:}\quad \frac{\partial \theta(x)}{\partial x} = \alpha, \tag{P1}$$

where $\alpha > 0$ is a constant. The forward pass is exact and binary; the backward pass deliberately overrides the true derivative, which is zero almost everywhere and undefined at zero.



(a) Exact binary forward pass.

(b) Constant backward signal.

Figure 4.1: The method uses an exact discrete neighborhood in the forward pass and a straight-through-style heuristic in the backward pass.

4.2 The $k + \epsilon$ neighborhood

Let $p(i)$ be the index of the k-th closest item to i, counting i itself. The paper defines

$$\mathbb{1}_{N_{k+\epsilon}}(i, j) = \theta(-D(i, j) + \operatorname{sg}(D(i, p(i))) + \epsilon)\,. \tag{P2}$$

Equivalently,

$$j \in N_{k+\epsilon}(i) \quad\Longleftrightarrow\quad D(i, j) \leq D(i, p(i)) + \epsilon.$$

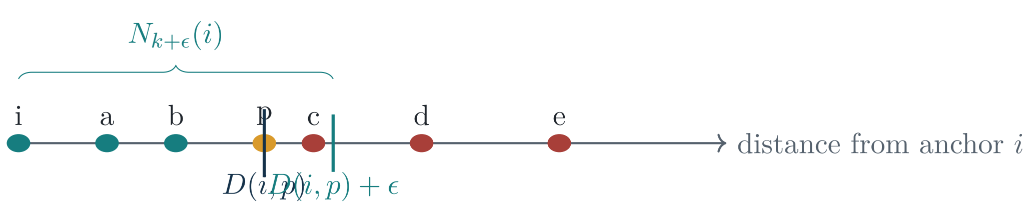


Figure 4.2: ϵ expands the boundary beyond the current k-th neighbor and acts as a ranking margin.

4.3 Why stop the gradient through the boundary?

The operator $\mathrm{sg}(\cdot)$ treats the k-th-neighbor distance as a constant during backpropagation. Otherwise, the optimization could alter the threshold itself instead of moving the candidate pair relative to the threshold. In code, this is the difference between

```
threshold.detach()
```

and an ordinary differentiable threshold.

```python
D = (2.0 - 2.0 * S).clamp_min(0.0)
# topk on -D returns smallest distances
neg_dk = (-D).topk(k, dim=1).values[:, -1:]
N = GreaterThan.apply(-D + eps, neg_dk.detach())
```

The mask N is binary in the forward pass. Its custom autograd function returns a constant gradient with opposite signs for the two comparison arguments.

4.4 Why not use a sigmoid?

A low-temperature sigmoid approximates a step, but its gradient becomes concentrated in a tiny interval around the boundary. A higher-temperature sigmoid spreads gradients but ceases to behave like a binary neighborhood. The paper reports that the exact-forward, constant-backward construction performs better in its ablations.

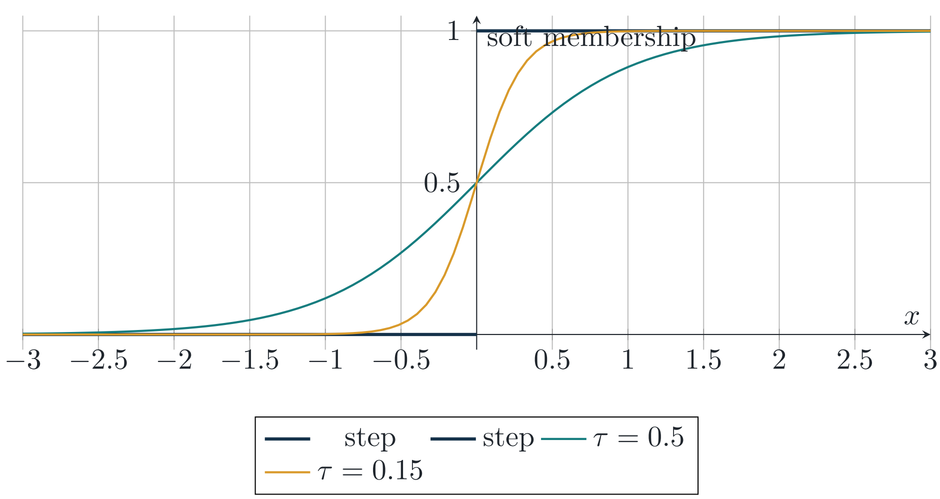


Figure 4.3: A sharper sigmoid is a better forward approximation but has useful gradients over a narrower region.

5

Step 2: Shared Neighborhood Context

5.1 Neighbor overlap as a dot product

Let $N_{ij} = \mathbb{1}_{N_{k+\epsilon}}(i,j)$. The i-th row N_i is a binary signature of the neighborhood around i. The number of shared neighbors is

$$M_+(i,j) = \sum_{p=1}^{n} N_{ip} N_{jp} = \langle N_i, N_j \rangle . \tag{5.1}$$

In matrix form,

$$M_+ = N N^\mathsf{T}. \tag{P22}$$

This is why the method can be implemented compactly with a matrix multiplication.

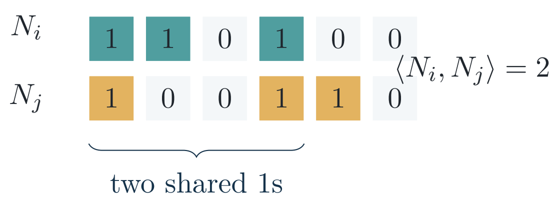


Figure 5.1: Binary neighborhood overlap is an ordinary dot product.

5.2 Why count shared non-neighbors?

Define $\bar{N} = 1 - N$. Then

$$M_-(i,j) = \sum_{p=1}^{n} (1 - N_{ip})(1 - N_{jp}) = \left\langle \bar{N}_i, \bar{N}_j \right\rangle . \tag{5.2}$$

Two neighborhood signatures can be similar because they include the same points *and* exclude the same points. The M_- term also prevents gradients from focusing only on entries where both neighborhood indicators are one.

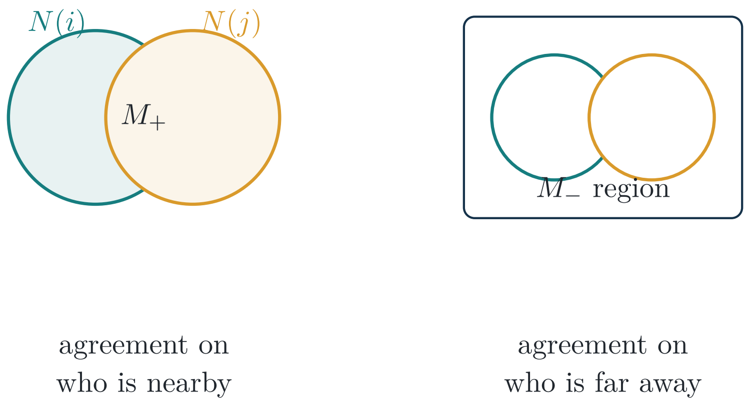


Figure 5.2: M_+ measures shared inclusion; M_- measures shared exclusion.

5.3 The intermediate contextual score W_1

The paper combines the two agreements, normalizes them, and gates the result by whether j is currently a neighbor of i:

$$\boxed{W_1(i,j) = \frac{N_{ij}}{2}\left(\frac{M_+(i,j)}{\mathrm{sg}\sum_p N_{ip}} + \frac{M_-(i,j)}{\mathrm{sg}\sum_p(1-N_{ip})}\right).} \tag{P3}$$

The three design choices are distinct:

1. **Gate N_{ij}:** contextual similarity is initially nonzero only when j belongs to i's neighborhood.
2. **Average M_+ and M_-:** reward agreement about both local inclusion and exclusion.
3. **Stop-gradient denominators:** normalize the score without training the model to manipulate neighborhood cardinality.

5.4 Algebra when $\epsilon = 0$

When $\epsilon = 0$, every row has exactly k ones, so $\sum_p N_{ip} = k$ and $\sum_p(1-N_{ip}) = n-k$. The complement overlap simplifies:

$$\begin{aligned} M_-(i,j) &= \sum_p (1-N_{ip})(1-N_{jp}) \\ &= n - 2k + \langle N_i, N_j \rangle . \end{aligned} \tag{P12}$$

Substitution gives

$$W_1(i,j) = N_{ij}\left(a\langle N_i, N_j\rangle + b\right), \tag{P14}$$

where

$$a = \frac{1}{2k} + \frac{1}{2(n-k)}, \qquad b = \frac{n-2k}{2(n-k)}. \tag{5.3}$$

Under the proposition's condition $n > 2k$, both a and b are positive. Thus, if $N_{ij} = 1$, then $W_1(i,j) > 0$ even before considering how large the overlap is.

Graph interpretation. Each row N_i is an adjacency vector in a directed k-nearest-neighbor graph. The dot product $\langle N_i, N_j \rangle$ counts common outgoing neighbors, so W_1 compares the *local graph signatures* of i and j, not merely their direct edge weight s_{ij}.

6

Step 3: Reciprocal Query Expansion and Symmetry

6.1 Reciprocal close neighbors

The method recomputes a smaller $k/2+\epsilon$ neighborhood and keeps only reciprocal relationships:

$$R_{k/2+\epsilon}(i,j) = N_{k/2+\epsilon}(i,j)N_{k/2+\epsilon}(j,i). \tag{P4a}$$

A one-sided accidental neighbor is rejected. A mutual neighbor is retained.

Figure 6.1: Reciprocity filters asymmetric nearest-neighbor accidents.

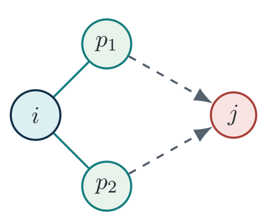
6.2 Query expansion

The method then averages W_1 over the reciprocal neighbors of i:

$$W_2(i,j) = \frac{\sum_p R_{k/2+\epsilon}(i,p)W_1(p,j)}{\sum_p R_{k/2+\epsilon}(i,p)}. \tag{6.1}$$

Finally it symmetrizes:

$$\boxed{w_{ij} = \frac{1}{2}\left(W_2(i,j) + W_2(j,i)\right).} \tag{P4}$$



Statistical intuition. Query expansion resembles local smoothing: a noisy pairwise estimate is replaced by an average across a small trusted neighborhood. Averaging can reduce variance, while reciprocity limits contamination from unreliable neighbors.

7

The Complete Objective and the Role of Every Term

7.1 Contextual loss

The final contextual matrix $W = [w_{ij}]$ is trained to match the label-similarity matrix Y:

$$\boxed{\mathcal{L}_{\text{context}} = \frac{1}{n^2} \sum_{i \neq j} (y_{ij} - w_{ij})^2.} \tag{P5}$$

The diagonal is excluded because every sample is trivially identical to itself.

The derivative with respect to a contextual score is ordinary MSE:

$$\frac{\partial \mathcal{L}_{\text{context}}}{\partial w_{ij}} = \frac{2}{n^2} (w_{ij} - y_{ij}). \tag{7.1}$$

The unusual part is the chain from w_{ij} back through query expansion, neighborhood intersections, and discrete threshold comparisons to the embedding similarities.

7.2 Similarity regularizer

The paper also regulates the mean pairwise cosine similarity:

$$\mathcal{L}_{\text{reg}} = \left(\tilde{s} - \frac{1}{n^2} \sum_{i,j} s_{ij} \right)^2. \tag{P7}$$

If all embeddings crowd into a small angular region, their mean similarity becomes too high. This term encourages broader use of the unit sphere.

7.3 The full objective

$$\boxed{\mathcal{L}_{\text{ours}} = \lambda \mathcal{L}_{\text{context}} + (1 - \lambda) \mathcal{L}_{\text{contrast}} + \gamma \mathcal{L}_{\text{reg}}.} \tag{P6}$$

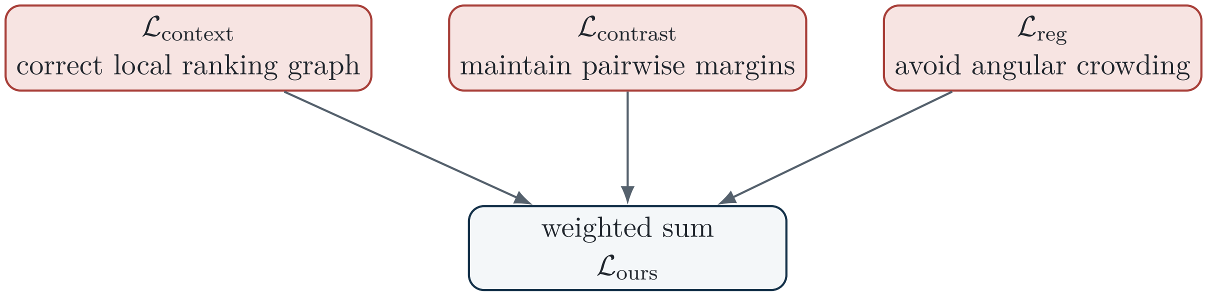


Figure 7.1: The three terms solve different failure modes.

7.4 The decomposability gap

A batch can be perfectly ranked even while the entire dataset is not. Once $\mathcal{L}_{\text{context}} = 0$ for an easy batch, that batch provides no further signal. The pairwise contrastive term continues enforcing absolute margins and helps bridge this difference between batch-wise and dataset-wise ranking.

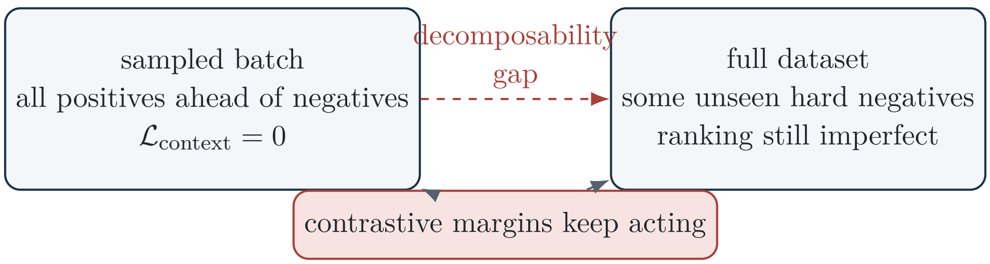


Figure 7.2: Why the hybrid objective is stronger than contextual loss alone.

7.5 Hyperparameter map

Parameter	Mathematical role	Practical interpretation
k	neighbor count	must equal samples per class in the balanced batch under the paper's theory
ϵ	expands the k-th-neighbor threshold	a flexible ranking margin in squared-distance units
α	surrogate derivative of θ	scales gradients through comparisons; largely redundant with learning rate
δ_+	positive similarity margin	positives below it are pulled closer
δ_-	negative similarity margin	negatives above it are pushed apart
λ	contextual–contrastive mixture	1 means contextual only; 0 means contrastive only
γ	similarity-regularizer weight	controls pressure to use more of the sphere
$\tilde{s}$	desired mean cosine similarity	target average angular spread

8

Why Zero Contextual Loss Means Correct Ranking

8.1 The paper's proposition

Proposition 8.1 (Liao–Tsiligkaridis–Kulis, Proposition 4.1). *Consider a balanced batch of size n with exactly k samples from each class, n divisible by k, $n > 2k$, $k \geq 2$, and $\epsilon = 0$. Then*

$$\mathcal{L}_{\text{context}} = 0$$

if and only if every positive sample is ranked ahead of every negative sample for every anchor:

$$s_{ip} > s_{ij} \quad \textit{whenever} \quad y_{ip} = 1, \; y_{ij} = 0.$$

Plain-language translation

Under the balanced-batch assumptions, the contextual loss has the right global minimum: it is zero exactly when every anchor's same-class examples occupy the full top-k neighborhood and every different-class example lies below them.

8.2 Forward direction: correct ranking implies zero loss

Assume every sample is correctly ranked.

Step 1. Each class contributes exactly k examples. Therefore the top-k neighborhood of anchor i is exactly its class:

$$N_{ij} = y_{ij}.$$

Step 2. If $N_{ij} = 1$, then i and j have identical neighborhoods of size k, so

$$\langle N_i, N_j \rangle = k.$$

Step 3. Substituting into the simplified W_1 formula yields $W_1(i, j) = N_{ij} = y_{ij}$.

Step 4. Reciprocal query expansion averages identical rows inside the same class, so it leaves these values unchanged.

Step 5. Hence $w_{ij} = y_{ij}$ for all off-diagonal pairs and the MSE is zero.

8.3 Backward direction: zero loss implies correct ranking

Assume $\mathcal{L}_{\text{context}} = 0$ but ranking is not correct. Because the batch contains exactly k positives for each anchor, including itself, a missing positive from the top-k set forces at least one negative intruder into the top-k set. So there exists a negative pair (i, j) with

$$y_{ij} = 0, \qquad N_{ij} = 1.$$

From equation (P14), $a > 0$ and $b > 0$, hence

$$W_1(i, j) = a \langle N_i, N_j \rangle + b > 0.$$

Query expansion is an average of nonnegative quantities with positive weights, so the corresponding contextual value remains positive:

$$w_{ij} > 0.$$

But $y_{ij} = 0$, so $(y_{ij} - w_{ij})^2 > 0$, contradicting zero loss. Therefore ranking must be correct.

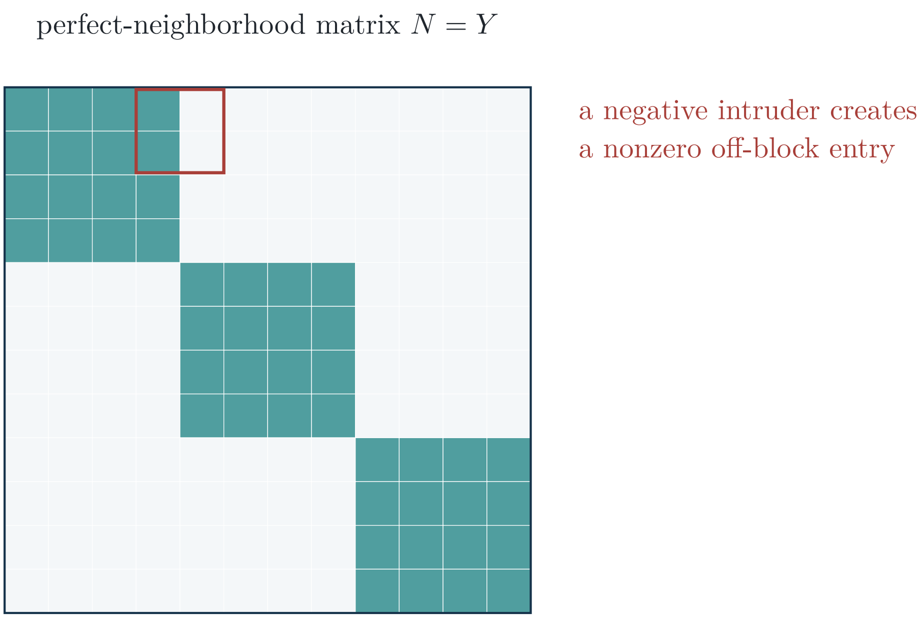


Figure 8.1: At the optimum, the neighbor graph has exactly the same block structure as label agreement.

8.4 What the assumptions are doing

Assumption	Why it matters
balanced k samples per class	makes the desired top-k set exactly equal to the same-label set
$\epsilon = 0$	fixes each neighborhood cardinality at exactly k
$n > 2k$	ensures the constant $b = (n - 2k)/(2(n - k))$ is positive in the proof
self included	makes a class of k examples correspond to a neighborhood of size k
outside these assumptions	the loss may remain useful empirically, but the proposition's zero-loss/correct-ranking equivalence is not guaranteed

9

Gradient Intuition and Semantic Consistency

9.1 Simplified gradient view

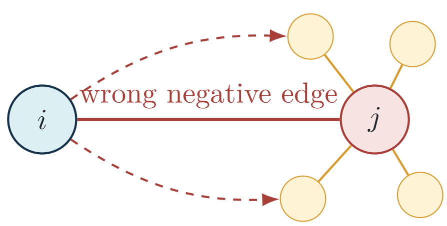
For $\epsilon = 0$, the paper writes

$$W_1(i,j) = N_{ij}\left(a\langle N_i, N_j\rangle + b\right). \tag{P8}$$

Consider a negative intruder $j \in N(i)$, so $y_{ij} = 0$ but $N_{ij} = 1$. The MSE wants to reduce $W_1(i,j)$. It can do so in two ways:

1. remove the direct neighborhood edge $i \to j$;
2. reduce overlap between the neighborhood signatures N_i and N_j.

The second route changes distances between i and members of j's neighborhood. That is the key distinction from a purely pairwise loss.



contextual gradient separates i from the *context* around j, not only from j

Figure 9.1: A contextual error propagates through a neighborhood.

9.2 Why apparently "wrong" pairwise gradients can generalize

The paper observes that a contextual gradient may sometimes pull apart same-label examples or push together different-label examples. Pairwise supervision would call that wrong. Contextual supervision asks a different question: is the binary label relation consistent with the local semantic structure?

The paper's gradient-clamping experiment enforces the pairwise label signs

$$\partial D(i,j) \leftarrow \begin{cases} \max(\partial D(i,j), 0), & y_{ij} = 1, \\ \min(\partial D(i,j), 0), & y_{ij} = 0, \end{cases} \tag{P10}$$

and reports higher train retrieval but lower test retrieval. Its interpretation is that some label-inconsistent gradients act as regularization against semantically inconsistent labels.

This argument concerns noisy or semantically incomplete *labels*. Pose-estimation corruption is noisy *input*. A corrupted skeleton may move an entire local neighborhood coherently or destroy it. Therefore the claim "context helps corrupted poses" remains an empirical hypothesis.

9.3 Stepwise gradient interpretation

The supplementary analysis can be summarized as follows:

Component	Positive pair error	Negative pair error
Step 1: N_{ij}	positive outside top-k is pulled inward	negative inside top-k is pushed outward
Step 2: overlap	align the two local contexts	separate each sample from the other's local context
Step 3: query expansion	smooth toward trusted-neighbor consensus	suppress one-sided accidental relations

10

A Worked Numerical Example

10.1 Setup

Use a balanced batch with $n = 12$, three classes, and $k = 4$ samples per class:

$$A = \{1, 2, 3, 4\}, \quad B = \{5, 6, 7, 8\}, \quad C = \{9, 10, 11, 12\}.$$

Suppose anchor 1 has a damaged neighborhood

$$N(1) = \{1, 2, 3, 5\},$$

so positive 4 is missing and negative 5 is an intruder. Let

$$N(2) = \{2, 1, 3, 4\}, \qquad N(5) = \{5, 6, 7, 1\}.$$

10.2 A true positive pair: $(1, 2)$

The shared neighbors are

$$N(1) \cap N(2) = \{1, 2, 3\}, \qquad M_+(1, 2) = 3.$$

The union contains five items, so seven of the twelve items are excluded by both neighborhoods:

$$M_-(1, 2) = 12 - |N(1) \cup N(2)| = 12 - 5 = 7.$$

Because $2 \in N(1)$, the gate is one. Therefore

$$W_1(1, 2) = \frac{1}{2}\left(\frac{3}{4} + \frac{7}{8}\right) \tag{10.1}$$

$$= \frac{13}{16} = 0.8125. \tag{10.2}$$

The target is $y_{12} = 1$, so this pair contributes

$$(1 - 0.8125)^2 \approx 0.0352$$

before the global $1/n^2$ factor.

10.3 A wrong negative pair: $(1, 5)$

Here

$$N(1) \cap N(5) = \{1, 5\}, \qquad M_+(1, 5) = 2.$$

The union has six items, so $M_{-}(1,5) = 6$. Since $5 \in N(1)$,

$$W_1(1,5) = \frac{1}{2}\left(\frac{2}{4} + \frac{6}{8}\right) \tag{10.3}$$

$$= 0.625. \tag{10.4}$$

But $y_{15} = 0$, so the contribution is

$$(0 - 0.625)^2 = 0.390625.$$

This is a strong signal to remove the intruder and separate the surrounding contexts.

10.4 A missing positive pair: $(1,4)$

Because $4 \notin N(1)$, the gate makes

$$W_1(1,4) = 0$$

even though $y_{14} = 1$. Its squared error is 1, the largest possible. Step 1 therefore pulls sample 4 toward the top-k boundary of anchor 1.

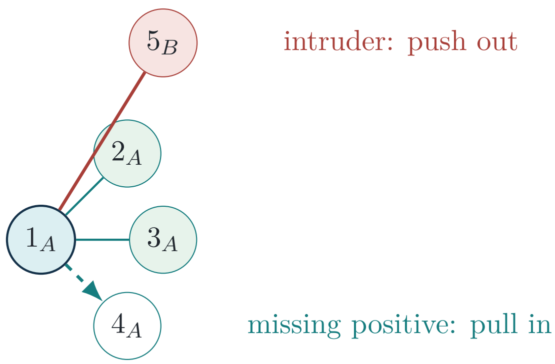


Figure 10.1: The numerical example as a neighborhood-repair problem.

10.5 What query expansion would do

Suppose 1 and 2 are reciprocal close neighbors. Then $W_2(1,j)$ averages the row of W_1 for anchors 1 and 2. Since anchor 2 has the correct neighbor 4 and excludes intruder 5, its row supplies a corrective vote: $W_2(1,4)$ rises and $W_2(1,5)$ falls. This is the local-consensus interpretation of query expansion.

11

Adapting the Mathematics to Pose Sequences

11.1 The input tensor

A pose sequence can be written

$$x \in \mathbb{R}^{T \times J \times C}, \tag{11.1}$$

where T is frames, J is joints, and $C \in \{2, 3\}$ is coordinate dimension. A sequence encoder produces one vector:

$$h = f_\theta(x, m), \qquad z = \frac{g_\phi(h)}{\|g_\phi(h)\|_2}. \tag{11.2}$$

Here m can be a validity mask for missing joints or frames, f_θ is MotionBERT or another temporal encoder, and g_ϕ is a small projection head.

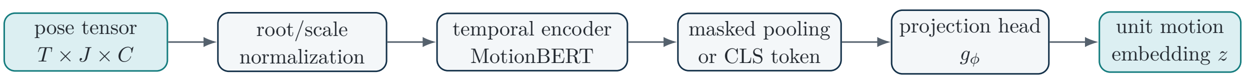


Figure 11.1: The loss does not care whether z came from an image or a motion encoder. Only the encoder-side computation changes.

11.2 Pose normalization

For joint j at frame t, let r be a root joint and b a body-scale statistic such as mean bone length. A simple normalization is

$$\tilde{x}_{t,j} = \frac{x_{t,j} - x_{t,r}}{b + \eta}, \tag{11.3}$$

with small $\eta > 0$ for stability. This removes global translation and much of body-size variation. View normalization, if used, should be treated as a separate design choice because it can remove information or create artifacts.

11.3 Controlled corruption operators

The main experiment corrupts the query and leaves the gallery clean. Let corruption severity be c.

11.3.1 Gaussian joint jitter

$$\tilde{x}_{t,j} = x_{t,j} + \sigma(c)\xi_{t,j}, \qquad \xi_{t,j} \sim \mathcal{N}(0, I_C). \tag{11.4}$$

Use scale-normalized coordinates so σ has comparable meaning across subjects.

11.3.2 Joint or limb dropout

Let $m_{t,j} \sim \text{Bernoulli}(1 - p(c))$. Then

$$\tilde{x}_{t,j} = m_{t,j} x_{t,j}, \tag{11.5}$$

while a companion mask tells the encoder which zeros are missing observations. Limb dropout uses a correlated mask over an anatomically connected subset.

11.3.3 Frame dropout

With frame mask $u_t \sim \text{Bernoulli}(1 - q(c))$,

$$\tilde{x}_t = u_t x_t, \tag{11.6}$$

or remove the frame and resample the remaining sequence to length T.

11.3.4 View rotation in 3D

For a yaw angle $\psi(c)$,

$$\tilde{x}_{t,j} = R_y(\psi) x_{t,j}, \qquad R_y(\psi) = \begin{bmatrix} \cos\psi & 0 & \sin\psi \\ 0 & 1 & 0 \\ -\sin\psi & 0 & \cos\psi \end{bmatrix}. \tag{11.7}$$

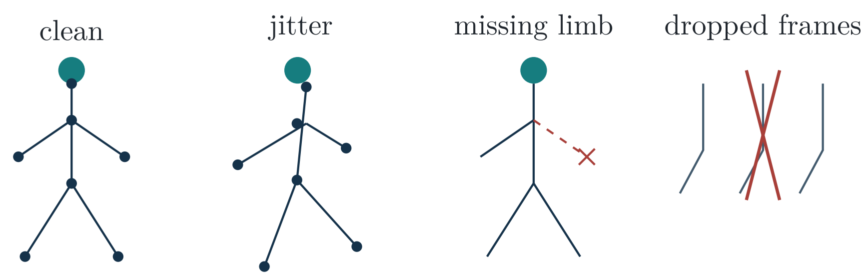


Figure 11.2: Schematic pose corruptions. These are controlled test interventions, not claims that one synthetic operator fully represents a real pose estimator.

11.4 The central hypothesis in mathematical form

Let $M_\ell(c)$ be a retrieval metric for loss ℓ at corruption severity c. Define degradation

$$\Delta_\ell(c) = M_\ell(0) - M_\ell(c). \tag{11.8}$$

The proposed hypothesis is not merely that contextual loss has larger clean $M(0)$, but that

$$\Delta_{\text{context}}(c) < \Delta_{\text{baseline}}(c) \quad \text{over a meaningful severity range.} \tag{11.9}$$

12

Implementation Blueprint

12.1 Recommended staged implementation

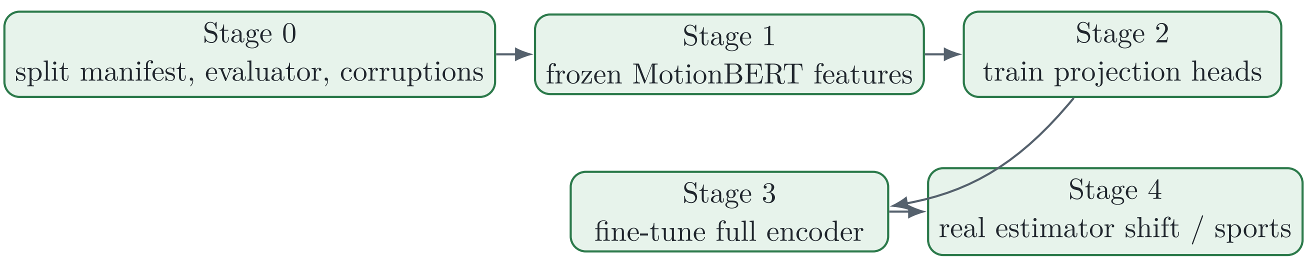


Figure 12.1: A low-risk sequence that isolates bugs and scientific effects.

1. **Freeze the encoder first.** Validate query/gallery construction, cosine ranking, metrics, and deterministic corruption curves.
2. **Train only a projection head.** Compare losses while holding the representation backbone fixed.
3. **Fine-tune the encoder only after stability.** This prevents an encoder bug or optimizer difference from masquerading as a loss effect.
4. **Add a corruption-aware baseline.** Compare ordinary SupCon with and without the same corruption augmentation, then add a MaskCLR-style control if feasible.

12.2 Reference PyTorch implementation

The following implementation follows the paper's equations, parameterizes k, uses exact forward masks with custom backward gradients, and handles empty contrastive sets safely.

```
from __future__ import annotations

from dataclasses import dataclass
from typing import Dict, Tuple

import torch
import torch.nn as nn
import torch.nn.functional as F


class GreaterThanSTE(torch.autograd.Function):
    """Exact >= in forward; constant surrogate gradient in backward."""

    @staticmethod
    def forward(ctx, x: torch.Tensor, y: torch.Tensor, alpha: float):
```

```python
        ctx.alpha = float(alpha)
        return (x >= y).to(dtype=x.dtype)

    @staticmethod
    def backward(ctx, grad_output: torch.Tensor):
        a = ctx.alpha
        return grad_output * a, -grad_output * a, None


def contextual_similarity(
    embeddings: torch.Tensor,
    *,
    k: int,
    eps: float,
    alpha: float = 10.0,
) -> Tuple[torch.Tensor, Dict[str, torch.Tensor]]:
    """Compute W from normalized or unnormalized [n, d] embeddings."""
    if embeddings.ndim != 2:
        raise ValueError("embeddings must have shape [batch, dim]")
    n = embeddings.shape[0]
    if k < 2 or k > n:
        raise ValueError(f"k must be in [2, {n}], got {k}")
    if k % 2 != 0:
        raise ValueError("k must be even for the k/2 reciprocal step")

    z = F.normalize(embeddings, p=2, dim=1)
    S = z @ z.T
    D = (2.0 - 2.0 * S).clamp_min(0.0)

    # Step 1: k+eps neighborhood.
    neg_topk = (-D).topk(k, dim=1).values
    neg_dk = neg_topk[:, -1:]
    N = GreaterThanSTE.apply(-D + eps, neg_dk.detach(), alpha)

    # Step 2: shared neighbors and shared non-neighbors.
    N_count = N.sum(dim=1, keepdim=True).detach().clamp_min(1.0)
    M_plus = (N @ N.T) / N_count

    N_bar = 1.0 - N
    N_bar_count = N_bar.sum(dim=1, keepdim=True).detach().clamp_min(1.0)
    M_minus = (N_bar @ N_bar.T) / N_bar_count

    W1 = 0.5 * (M_plus + M_minus) * N

    # Step 3: reciprocal k/2 neighborhood and query expansion.
    k2 = k // 2
    neg_dk2 = neg_topk[:, k2 - 1 : k2]
    N2 = GreaterThanSTE.apply(-D + eps, neg_dk2.detach(), alpha)
    R = N2 * N2.T
    R_count = R.sum(dim=1, keepdim=True).clamp_min(1.0)
    W2 = (R @ W1) / R_count
    W = 0.5 * (W2 + W2.T)

    aux = {"z": z, "S": S, "D": D, "N": N, "W1": W1, "R": R}
    return W, aux


@dataclass(frozen=True)
```

```python
class ContextualLossConfig:
    k: int
    eps: float = 0.05
    alpha: float = 10.0
    lam: float = 0.2
    gamma: float = 0.1
    target_mean_similarity: float = 0.0
    positive_margin: float = 0.9
    negative_margin: float = 0.6


class ContextualMetricLoss(nn.Module):
    def __init__(self, cfg: ContextualLossConfig):
        super().__init__()
        self.cfg = cfg

    @staticmethod
    def _mean_active(x: torch.Tensor) -> torch.Tensor:
        active = x > 0
        if not torch.any(active):
            return x.new_zeros(())
        return x[active].mean()

    def forward(self, embeddings: torch.Tensor, labels: torch.Tensor):
        W, aux = contextual_similarity(
            embeddings,
            k=self.cfg.k,
            eps=self.cfg.eps,
            alpha=self.cfg.alpha,
        )
        S = aux["S"]
        Y = labels[:, None].eq(labels[None, :]).to(S.dtype)
        off_diag = 1.0 - torch.eye(S.shape[0], device=S.device, dtype=S.dtype)

        L_context = (((W - Y) ** 2) * off_diag).mean()

        pos_hinge = F.relu(self.cfg.positive_margin - S) * Y * off_diag
        neg_hinge = F.relu(S - self.cfg.negative_margin) * (1.0 - Y)
        L_contrast = self._mean_active(pos_hinge) + self._mean_active(neg_hinge)

        L_reg = (S.mean() - self.cfg.target_mean_similarity).square()
        total = (
            self.cfg.lam * L_context
            + (1.0 - self.cfg.lam) * L_contrast
            + self.cfg.gamma * L_reg
        )
        stats = {
            "loss": total.detach(),
            "context": L_context.detach(),
            "contrast": L_contrast.detach(),
            "reg": L_reg.detach(),
            "mean_similarity": S.mean().detach(),
        }
        return total, stats
```

12.3 Training loop for pose sequences

```
for pose, valid_mask, label in train_loader:
    pose = pose.to(device) # [n, T, J, C]
    valid_mask = valid_mask.to(device)
    label = label.to(device)

    h = motion_encoder(pose, valid_mask=valid_mask)
    z = projection_head(h)

    loss, stats = metric_loss(z, label)

    optimizer.zero_grad(set_to_none=True)
    loss.backward()
    torch.nn.utils.clip_grad_norm_(parameters, max_norm=5.0)
    optimizer.step()
```

12.4 Sampler invariant

Every training batch should satisfy

$$\#\{i : y_i = c\} = k \quad \text{for every class } c \text{ represented in the batch.} \tag{12.1}$$

Use assertions during development:

```
unique, counts = torch.unique(labels, return_counts=True)
assert torch.all(counts == k), (unique, counts)
assert batch_size % k == 0
assert k % 2 == 0
```

12.5 Complexity and memory

The dominant contextual operations multiply $n \times n$ matrices, so arithmetic scales as $O(n^3)$ with batch size. The paper also describes empirical time and space scaling as cubic, while emphasizing that GPU matrix multiplication makes the observed cost practical at its studied batch sizes. For pose models, the encoder may dominate compute, but long sequences can make batch size the binding constraint. Profile rather than assume.

Practical levers:

- start with $16 \times 4 = 64$ or $32 \times 4 = 128$ batches;
- shorten or subsample sequences before shrinking k below four;
- use mixed precision for the encoder, but inspect neighborhood ties and numerical stability;
- gradient accumulation does *not* enlarge the contextual neighborhood, because each micro-batch forms a separate graph;
- parameterize k explicitly—the public 256-pixel reference path hard-codes $k = 4$.

Gradient accumulation increases effective optimizer batch size but does not reproduce a larger contextual batch. The loss only sees the current micro-batch's $n \times n$ graph. If neighborhood diversity matters, you need an actual larger forward batch, a memory-bank extension, or a changed method.

12.6 Unit tests before training

1. **Symmetry:** verify $W \approx W^\top$.
2. **Range:** verify $0 \leq W_{ij} \leq 1$ up to floating error.
3. **Perfect blocks:** construct embeddings with well-separated class clusters and check near-zero contextual loss.
4. **Broken pair:** move one negative into an anchor's top-k set and check the loss increases.
5. **Gradient existence:** call backward and verify finite, nonzero gradients.
6. **Sampler:** assert exact k counts per represented class.
7. **Deterministic corruptions:** fixed seed and sample ID must reproduce identical corrupted queries.

13

Retrieval Evaluation and a Fair Experimental Design

13.1 Query/gallery construction

Use disjoint query, gallery, and validation partitions. The query clip must not occur in the gallery. The primary deployment model is

$$\text{corrupted query} \quad \longrightarrow \quad \text{fixed clean gallery.}$$

A secondary experiment may corrupt both sides, but it answers a different question.

13.2 Recall at K

For query q, let $\pi_q(r)$ be the gallery item at rank r. Then

$$\mathrm{R@K}(q) = \mathbb{1}\left[\exists r \leq K : \mathrm{rel}(q, \pi_q(r)) = 1\right]. \tag{13.1}$$

Dataset R@K is the mean over queries.

13.3 Average precision and mAP

Let R_q be the number of relevant gallery items for q, and let $\mathrm{Prec}_q(r)$ be precision among the first r retrieved items. Then

$$\mathrm{AP}(q) = \frac{1}{R_q} \sum_{r=1}^{m} \mathrm{Prec}_q(r)\ \mathrm{rel}(q, \pi_q(r)), \tag{13.2}$$

with

$$\mathrm{mAP} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathrm{AP}(q). \tag{13.3}$$

R@1 asks whether the first answer is useful; mAP evaluates the ordering of all relevant examples.

13.4 Robustness curves

Report the raw metric and the clean-to-corrupted drop. A useful summary beyond the proposal is the normalized retention

$$\rho_\ell(c) = \frac{M_\ell(c)}{M_\ell(0) + \eta}. \tag{13.4}$$

This separates clean strength from robustness. It should supplement, not replace, absolute metrics.

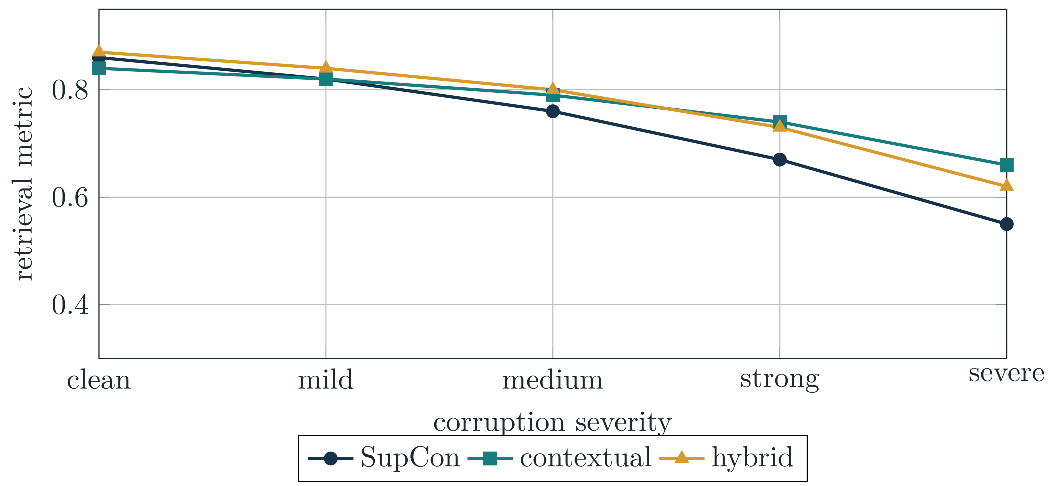


Figure 13.1: Schematic robustness curves only—not experimental results. The target result is a smaller degradation slope, not necessarily the highest clean point.

13.5 Core comparison matrix

Method	Same encoder?	Same corruption exposure?	Scientific role
cross-entropy embedding	yes	matched	classification baseline
triplet or multi-similarity	yes	matched	conventional metric learning
supervised contrastive	yes	matched	strong batch-contrastive baseline
contextual only	yes	matched	isolate neighborhood objective
hybrid contextual+contrastive	yes	matched	paper's complete mechanism
corruption-aware SupCon	yes	same augmented inputs	separate loss effect from augmentation
MaskCLR-style control	preferably yes	protocol-matched	robustness-specific comparator

13.6 Fairness controls

Following the metric-learning methodology cautions in Musgrave et al., hold fixed:

- encoder architecture and initialization;
- embedding dimension and normalization;
- data split and query/gallery manifest;
- augmentation and corruption exposure;
- number of updates and tuning budget;

- batch composition, except where a method inherently requires a different sampler;
- model-selection metric and validation set;
- reporting over multiple seeds.

13.7 Recommended experiment order

1. clean frozen-feature retrieval;
2. corruption curves with no training;
3. projection-head SupCon vs contextual vs hybrid;
4. full-encoder fine-tuning;
5. corruption-aware training control;
6. cross-subject and cross-setup replication;
7. optional pose-estimator shift;
8. optional SportsPose or fine-grained sports extension.

14

Discussion Preparation

14.1 A 60-second explanation

Use this almost verbatim

"I want to test whether contextual metric learning transfers from image retrieval to pose-sequence retrieval. A temporal encoder maps each skeleton sequence to a normalized vector, and cosine similarity ranks a clean gallery. Standard metric losses supervise individual pairs or triplets. The contextual objective also compares their local neighborhoods: it builds a differentiable top-k graph, measures shared and reciprocal neighbors, and trains that graph to match label agreement. The original paper shows robustness to label noise, but that does not imply robustness to corrupted pose coordinates. My experiment isolates that question by holding the encoder, data split, batch composition, and corruption protocol fixed while comparing supervised contrastive, contextual, and hybrid losses on clean and increasingly corrupted queries."

14.2 The five equations to narrate

1. $D(i,j) = 2 - 2s_{ij}$: normalized cosine geometry.
2. $N_{ij} = \theta(-D_{ij} + \mathrm{sg}(D_{i,k}) + \epsilon)$: differentiable neighborhood membership.
3. W_1: normalized agreement on shared neighbors and shared non-neighbors.
4. W_2 and W: reciprocal query expansion and symmetrization.
5. $\mathcal{L} = \lambda\mathcal{L}_{context} + (1-\lambda)\mathcal{L}_{contrast} + \gamma\mathcal{L}_{reg}$: ranking, margins, and space usage.

14.3 Likely questions and precise answers

Question	Answer
What is the novelty?	Not pose retrieval itself and not a new encoder. The novelty is a controlled test of whether a supervised neighborhood-ranking objective improves *retrieval robustness to corrupted pose inputs*.
Why should context help input noise?	It may reduce reliance on one unstable pairwise comparison by aggregating local neighborhood evidence. That is a hypothesis; coherent neighborhood shifts could also make it fail.

Why NTU120 first?	It provides scale, many action classes, standard subject/setup protocols, and compatibility with a public motion-embedding pipeline. It is an engineering and methodology benchmark, not proof of fine-grained technique understanding.
Why k equals samples per class?	Under balanced sampling, perfect top-k neighborhoods then equal the same-label set. This is the condition behind the ranking proposition.
Why not only use SupCon?	SupCon is the strongest simple baseline. The project asks whether explicit neighborhood overlap adds anything beyond many-positive contrastive learning.
What is the cleanest primary endpoint?	The difference in degradation curves under query corruption, with the gallery fixed and clean, reported using R@1/5/10 and mAP.
What would a null result mean?	Robustness comes mainly from encoder pretraining or corruption exposure rather than the neighborhood objective. That is still a useful separation of mechanisms.
What is the main technical risk?	Batch-level cubic work and the need for enough samples per class. Profile 16×4 and 32×4 batches before full experiments.
Could action-class relevance be too coarse?	Yes. It is appropriate for the first falsifiable benchmark, but it cannot support claims about movement quality, technique, or rehabilitation status.
How will you ensure fairness?	Same backbone, initialization, split, sampler, augmentation, tuning budget, and evaluator; only the objective changes in the core comparison.

14.4 Three decisions to ask the professor

1. Is NTU class-level retrieval an appropriate first scientific target, or should relevance be fine-grained from the outset?

2. Should the first deliverable be a single-backbone loss comparison, or include a full MaskCLR-style robustness baseline immediately?

3. Is the strongest framing "contextual metric learning for temporal data" or "retrieval robustness under pose-estimation noise"?

14.5 Claims to avoid

- "Contextual loss is robust to pose noise." It has not yet been tested.
- "NTU retrieval measures technique quality." It measures class relevance under the chosen protocol.
- "The project invents a new motion architecture." The initial strength is controlled integration, not architecture novelty.
- "A higher clean R@1 proves robustness." Robustness requires degradation curves or cross-domain evaluation.

- “Gradient accumulation solves the batch-size problem.” It does not create a larger contextual graph.

14.6 Final mental model

One sentence per layer

1. **Geometry:** normalized vectors turn cosine ranking into distance ranking.
2. **Graph:** top-k comparisons turn the similarity matrix into a local directed graph.
3. **Context:** shared neighbors and non-neighbors compare graph signatures.
4. **Consensus:** reciprocal query expansion smooths scores through trusted neighbors.
5. **Learning:** MSE matches the contextual graph to label agreement, while contrastive margins and a mean-similarity regularizer stabilize the embedding.
6. **Research question:** does that graph-based inductive bias preserve motion retrieval when the observed skeleton is damaged?

A

Notation Reference

Symbol	Meaning
x_i	input item; an image in the foundation paper or pose sequence in the proposal
h_i	unnormalized encoder output
$z_i \in \mathbb{R}^d$	unit-normalized embedding
s_{ij}	cosine similarity $z_i^\intercal z_j$
$D(i,j)$	squared Euclidean distance $2 - 2s_{ij}$
n	mini-batch size
k	samples per class and contextual neighborhood size
y_{ij}	binary label similarity
θ	Heaviside comparison with custom backward gradient
α	constant surrogate derivative for θ
ϵ	neighborhood margin
$N_{k+\epsilon}(i)$	expanded k-nearest-neighbor set of sample i
M_+	count of shared neighbors
M_-	count of shared non-neighbors
W_1	initial gated contextual similarity
$R_{k/2+\epsilon}(i)$	reciprocal close-neighbor set
W_2	query-expanded contextual similarity
w_{ij}	final symmetric contextual similarity
δ_+, δ_-	positive and negative contrastive margins
$\tilde{s}$	target mean cosine similarity
λ, γ	loss-mixture weights

B

Detailed Algebra and Gradient Notes

B.1 Complement-intersection simplification

Starting from

$$M_-(i,j) = \sum_p (1 - N_{ip})(1 - N_{jp}),$$

expand:

$$\begin{aligned} M_-(i,j) &= \sum_p (1 - N_{ip} - N_{jp} + N_{ip}N_{jp}) && \text{(B.1)} \\ &= n - \sum_p N_{ip} - \sum_p N_{jp} + \sum_p N_{ip}N_{jp} && \text{(B.2)} \\ &= n - 2k + \langle N_i, N_j \rangle \,. && \text{(B.3)} \end{aligned}$$

Substituting into W_1 gives the constants a and b in equation (5.3).

B.2 Step-1 surrogate gradient

For the simplified loss

$$\mathcal{L}_1 = \mathrm{MSE}(y_{ij}, N_{ij}),$$

a negative sample inside $N(i)$ receives a constant push outward, while a positive sample outside $N(i)$ receives a constant pull inward. This resembles contrastive loss with a dynamic boundary: the current k-th-neighbor distance replaces a fixed margin.

B.3 Step-2 context gradient

For a wrongly included negative pair and $\epsilon = 0$, the supplementary derivation shows that optimizing neighborhood overlap changes the distance from i to every sample p depending on whether p lies inside j's neighborhood. In compact qualitative form,

$$\mathrm{sign}\left(\frac{\partial \mathcal{L}}{\partial D(i,p)}\right) = \begin{cases} -1, & p \in N(j), \\ +1, & p \notin N(j), \end{cases}$$

under the paper's sign convention for the considered negative error. Gradient descent therefore increases distance to j's context and decreases distance to points outside that context. The exact coefficients depend on k, $n-k$, α, and the upstream MSE gradient.

B.4 Logical AND as multiplication

The reciprocal mask uses

$$R = N_2 \odot N_2^{\mathsf{T}}.$$

For binary inputs, multiplication equals logical AND. On the continuous backward surrogate, multiplication is smooth but has zero derivative when the other factor is zero. The paper's ablation favors multiplication over a minimum-based alternative.

C

Implementation Checklist

Before the first real run

1. Freeze and version the query/gallery split manifest.
2. Verify no query ID appears in the gallery.
3. Save one deterministic corrupted copy per severity for debugging.
4. Confirm every batch has exactly k examples per represented class.
5. Log $\mathcal{L}_{context}$, $\mathcal{L}_{contrast}$, $\mathcal{L}_{reg}$, mean similarity, and gradient norm separately.
6. Visualize S, N, W_1, and W as heatmaps for a small batch.
7. Check clean retrieval before any corruption experiment.
8. Compare exact brute-force cosine retrieval against any FAISS index before using approximate search.
9. Tune each baseline on validation data with a matched budget; never select on test corruption curves.
10. Run at least three seeds once the pipeline is stable.

D

Two-Minute Detachable Cheat Sheet

Core proposition

Question: Does contextual neighborhood optimization preserve pose-sequence retrieval better than standard metric losses as skeleton observations become corrupted?
Design: same MotionBERT encoder, NTU120 split, batch sampler, augmentations, and evaluator; compare SupCon, contextual, and hybrid; corrupt query only against a fixed clean gallery.

Five equations

$$D = 2 - 2S$$

$$N_{ij} = \theta(-D_{ij} + \mathrm{sg}(D_{i,k}) + \epsilon)$$

$$W_1 = \frac{1}{2}\left(\frac{NN^\top}{|N_i|} + \frac{(1-N)(1-N)^\top}{|N_i^c|}\right) \odot N$$

$$W = \tfrac{1}{2}\left(\frac{RW_1}{|R_i|} + \left(\frac{RW_1}{|R_i|}\right)^\top\right)$$

$$\mathcal{L} = \lambda\,\mathrm{MSE}(W, Y) + (1-\lambda)\mathcal{L}_{contrast} + \gamma\mathcal{L}_{reg}$$

What makes the method non-standard

Exact binary top-k comparisons in the forward pass; constant heuristic gradients in the backward pass; neighborhood overlap and reciprocal query expansion; k tied to samples per class.

What remains unknown

The paper's label-noise robustness does not prove pose-input robustness. The project must measure corruption degradation curves and include corruption-aware controls.

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